STAFF REVIEW · H446 051 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. First-in-first-out, from front.

2. LIFO versus FIFO.

3. Value identifying an object instance.

4. Set node7.next = node5.next, then node5.next = node7. Saving the existing successor first preserves access to node9.

I Do — source worked steps

1. Explain the trace — New order A,X,B. The inserted node links between A and B.

We Do — guided CHECK

1. Explain the trace
P,R remain. After deletion P and R are adjacent logical neighbours.

You Do — source mark schemes

1. 3.next becomes 4 [2], 4.next=6 [2], final HEAD→3→4→6→NULL [2].

2. var l=new LinkedList<int>(); [1]; l.AddLast(2); l.AddLast(4); [2]; l.AddAfter(l.First!,3); [3].

3. List indexed access O(1) [2], insertion shifts subsequent values O(n) [1]; linked-list insertion with node reference O(1) links [2], finding node may take O(n) [1].

Exit Ticket — source answer

1. False. Traversal is generally required.